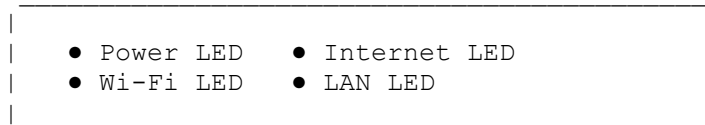


Routers and Switches Assignment

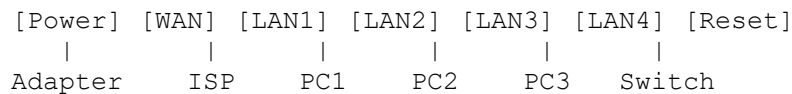
Task 1: Labeled Diagram of a Home Wi-Fi Router and Network Switch

Home Wi-Fi Router

HOME Wi-Fi ROUTER



Back View

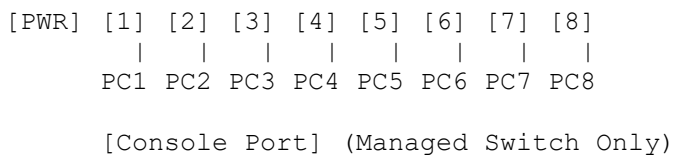


Labels

- Power Port – Connects the power adapter.
- WAN Port – Connects the Internet cable from the ISP.
- LAN Ports – Connect desktop PCs, switches, or printers.
- Status LEDs – Show power, Internet, Wi-Fi, and LAN activity.

Network Switch

NETWORK SWITCH



Labels

- Power Port
- Ethernet Ports (1–8)
- Console Port (Managed Switch)
- Power/Link LEDs

Task 2: Router vs Switch Comparison

Feature	Router	Switch
Main Function	Connects different networks (LAN to Internet).	Connects devices within the same LAN.
Address Used	IP Address	MAC Address
Traffic Handling	Routes packets between different networks.	Forwards frames to the correct device.
Internet Sharing	Yes	No
DHCP/NAT	Supported	Not normally supported

Three Key Differences

1. A router connects different networks, while a switch connects devices within the same network.
2. A router uses **IP addresses**, whereas a switch uses **MAC addresses**.
3. A router can provide Internet access, while a switch only expands the number of network connections.

Task 3: Identifying Router and Switch Components

Router

Power Port

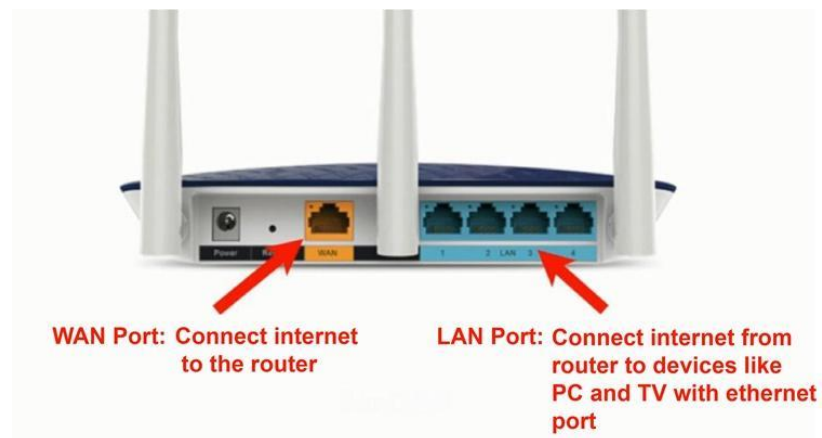
- Rear side of the router.
- Connects the power adapter.

WAN Port

- Usually a blue or yellow Ethernet port.
- Connects to the ISP modem.

LAN Ports

- Four Ethernet ports beside the WAN port.
- Used for connecting computers.



LED Indicators

- Located on the front panel.
- Display Power, Internet, Wi-Fi, and LAN status.

Network Switch

Power Port

- Located on one side of the switch.

Ethernet Ports

- Eight or more RJ-45 ports on the front.

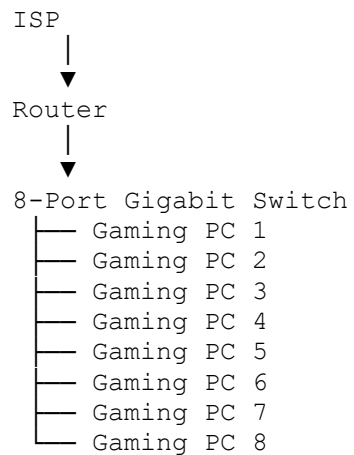
LED Indicators

- Small LEDs above each Ethernet port.
- Indicate power and network activity.

Task 4: Gaming Cafe Network Scenario

A small gaming café has an Internet connection from the ISP and eight gaming computers.

Network Setup



Explanation

The ISP cable is first connected to the **router** because it manages the Internet connection, assigns IP addresses (DHCP), and provides network security through NAT and the firewall.

The router is then connected to an **8-port Gigabit switch**, which expands the number of available Ethernet ports.

Each gaming PC is connected to the switch using Cat6 Ethernet cables. This setup provides high-speed, low-latency communication for multiplayer games and smooth IPL streaming.